

Data Preprocessing in Data Mining

Ensuring Data Quality for Superior Knowledge Discovery



DEFINING DATA PREPROCESSING

The Concept

Data Preprocessing is a crucial technique in data mining that involves transforming raw, "dirty" data into an understandable and clean format. It prepares datasets for analysis by removing noise and inconsistencies.

The Challenge

Real-world data is often incomplete, inconsistent, and contains errors. Without preprocessing, mining results are unreliable and computationally expensive to obtain.

| THE NEED FOR DATA QUALITY

- ✓ **Accuracy:** To remove errors or outliers that skew results.
- ✓ **Completeness:** To handle missing values in critical attributes.
- ✓ **Consistency:** To ensure uniform naming conventions (e.g., "M" vs "Male").
- ✓ **Efficiency:** Reducing the computational load on mining algorithms.
- ✓ **Believability:** Trusting the output of a mining model.

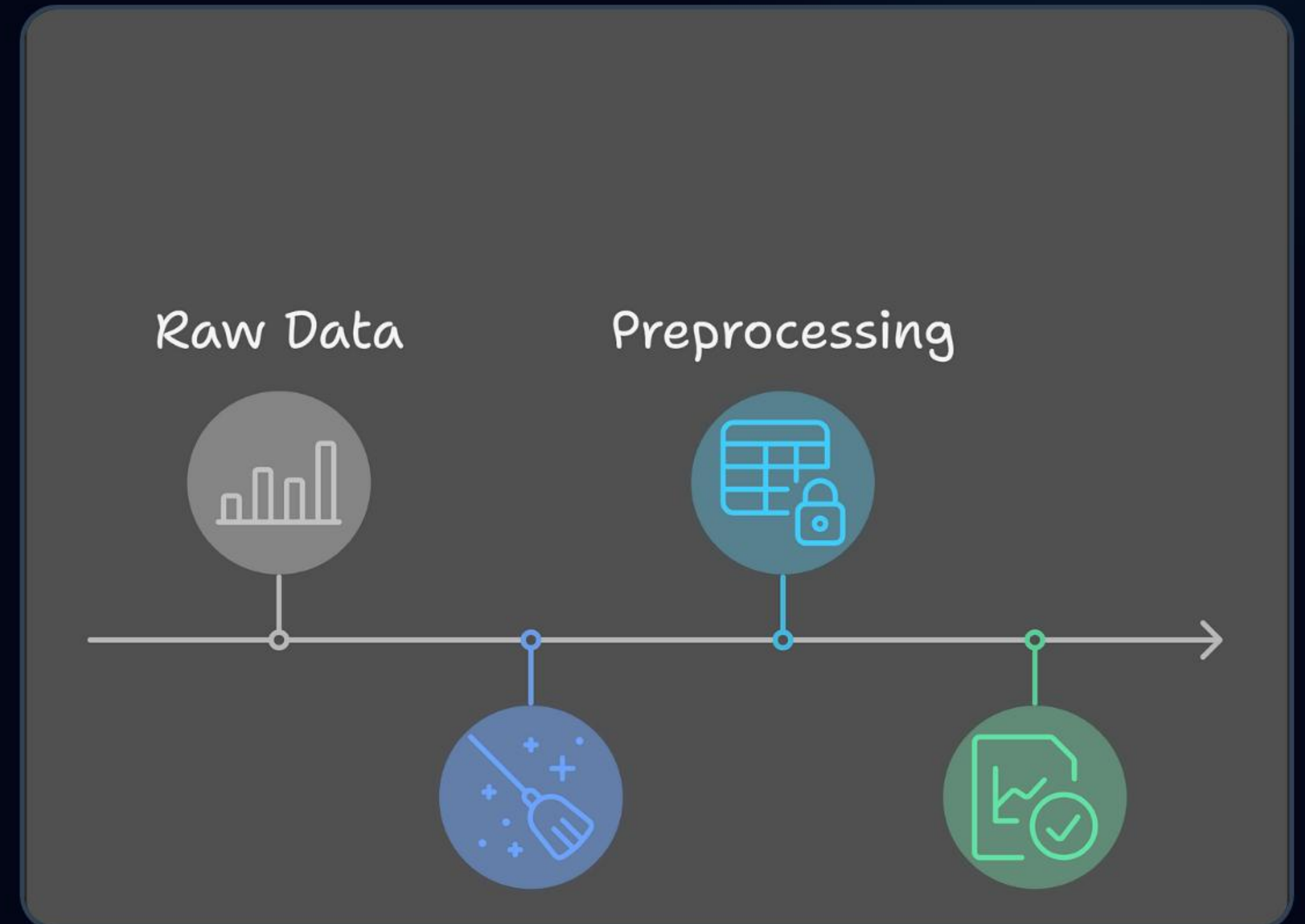
| 1. DATA CLEANING

Cleaning handles "dirty" data—the most time-consuming step in preprocessing.

Missing Values: Ignore tuples or use global constants/mean imputation.

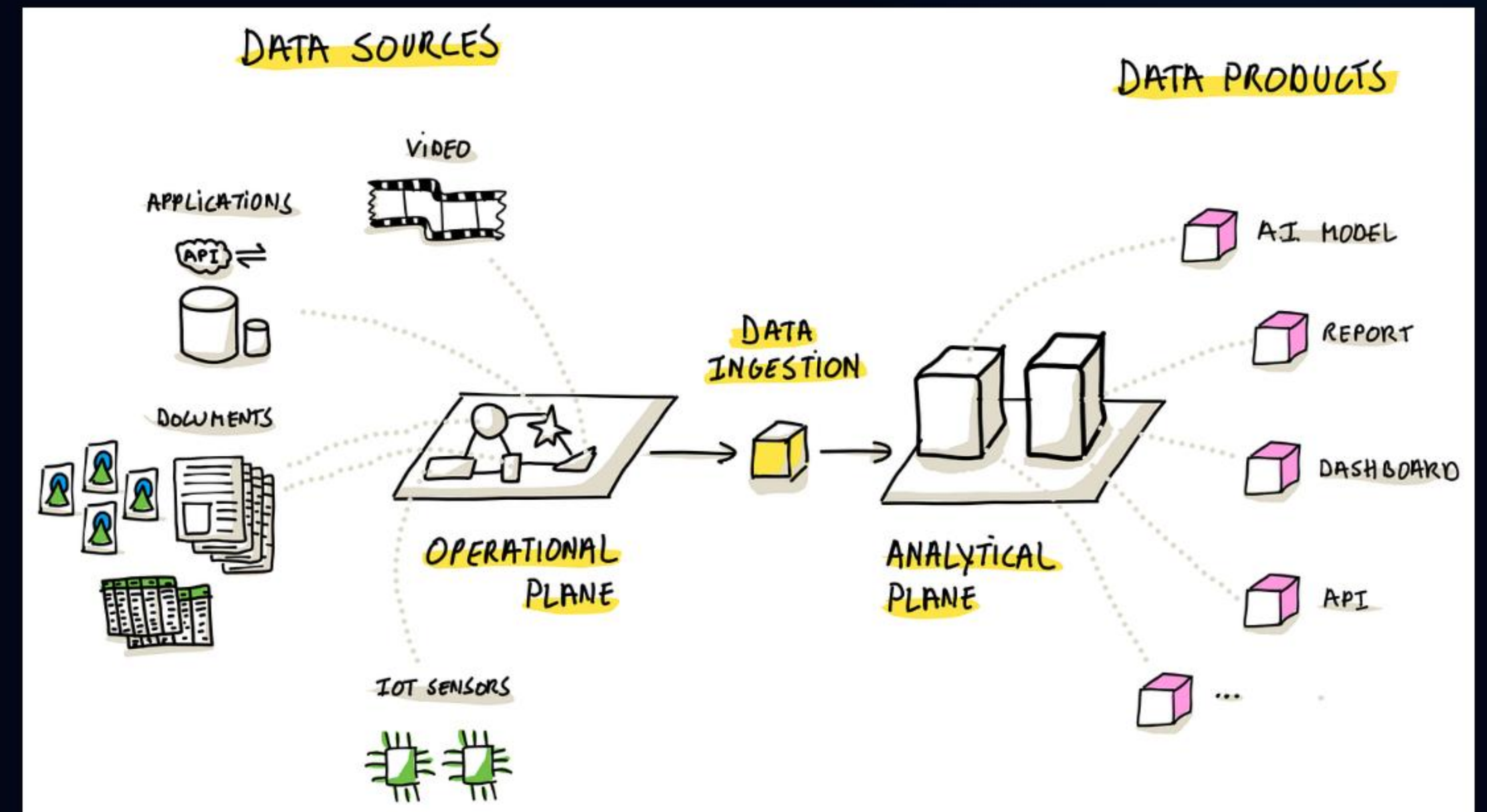
Noisy Data: Use Binning, Regression, or Clustering to smooth noise.

Inconsistencies: Fix mismatched labels or structural errors.



2. DATA INTEGRATION

-  **Schema Integration:** Mapping entities from multiple databases.
-  **Entity Identification:** Recognizing that "Cust_ID" and "User_Number" refer to the same object.
-  **Redundancy Control:** Removing attributes that can be derived from others (e.g., Age vs DoB).



| 3. DATA TRANSFORMATION

Transforming data into formats suitable for mining algorithms.

Method	Description	Key Application
Normalization	Scaling values to a small, specified range (e.g., 0 to 1).	Distance-based algorithms.
Aggregation	Summarizing data (e.g., daily sales to monthly).	Trend analysis.
Generalization	Replacing low-level data with high-level concepts.	Data reduction/Privacy.
Encoding	Converting categorical data into numeric values.	Neural networks.

Min-Max Normalization Formula:

$$v' = \frac{v - \min_A}{\max_A - \min_A} (\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A$$

| 4. DATA REDUCTION

60%

Reducing Volume

Data reduction aims to produce a smaller representation of the data that yields the same analytical results. This significantly reduces computation time and cost.



Compression

Using algorithms like Wavelet transforms.



Dimensionality

Removing irrelevant attributes (PCA).



Numerosity

Histograms, sampling, and clustering.

PREPROCESSING FLOWCHART



Figure: The Knowledge Discovery in Databases (KDD) Pipeline.

| PROS AND CONS

Advantages

- Higher prediction accuracy.
- Faster training and processing times.
- Reduction in noise and model bias.
- Easier data visualization.

Limitations

- Resource intensive (CPU/Memory).
- Potential loss of critical information.
- Domain expertise is required.
- Time-consuming (80% of project time).

CASE STUDY

Fraud Detection in Banking

A global bank processing 10 million transactions daily faced massive data noise from network glitches and redundant records.

- ✓ **Cleaning:** Removed incomplete sessions.
- ✓ **Reduction:** Focused on high-risk geography attributes.
- ✓ **Result:** 30% increase in fraud detection accuracy.



| PRACTICAL APPLICATIONS



Healthcare

Normalizing patient vitals from diverse sensors for diagnostics.



E-Commerce

Cleaning purchase histories to improve recommendation engines.



Manufacturing

Reducing sensor data volume for real-time predictive maintenance.

Thank You

Any Questions on Data Preprocessing?

References

- Han, J., Kamber, M., & Pei, J. (2011). Data Mining: Concepts and Techniques.
- GeeksforGeeks: Data Preprocessing in Data Mining.
- Meegle Tutorial: Quality Data Mining Framework.
- Denodo: Guide to Feature Scaling and Reduction.